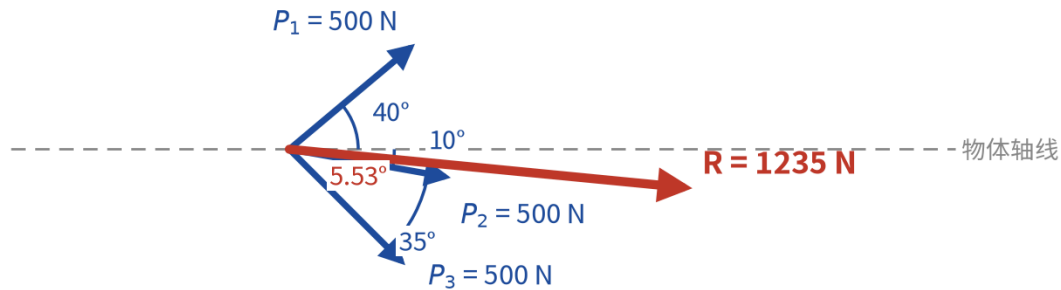
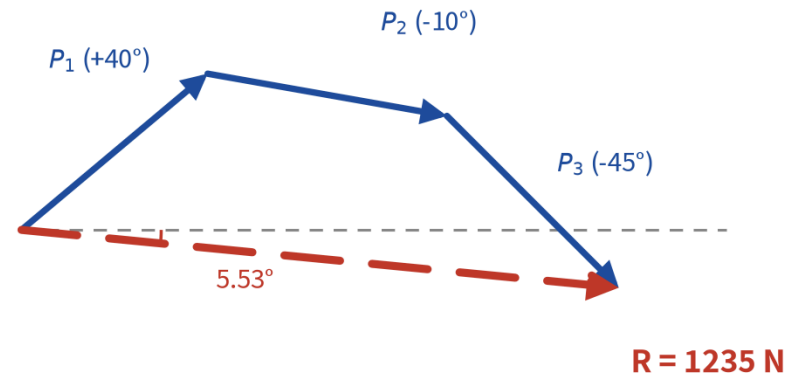


(a) 三力合成：三力与合力



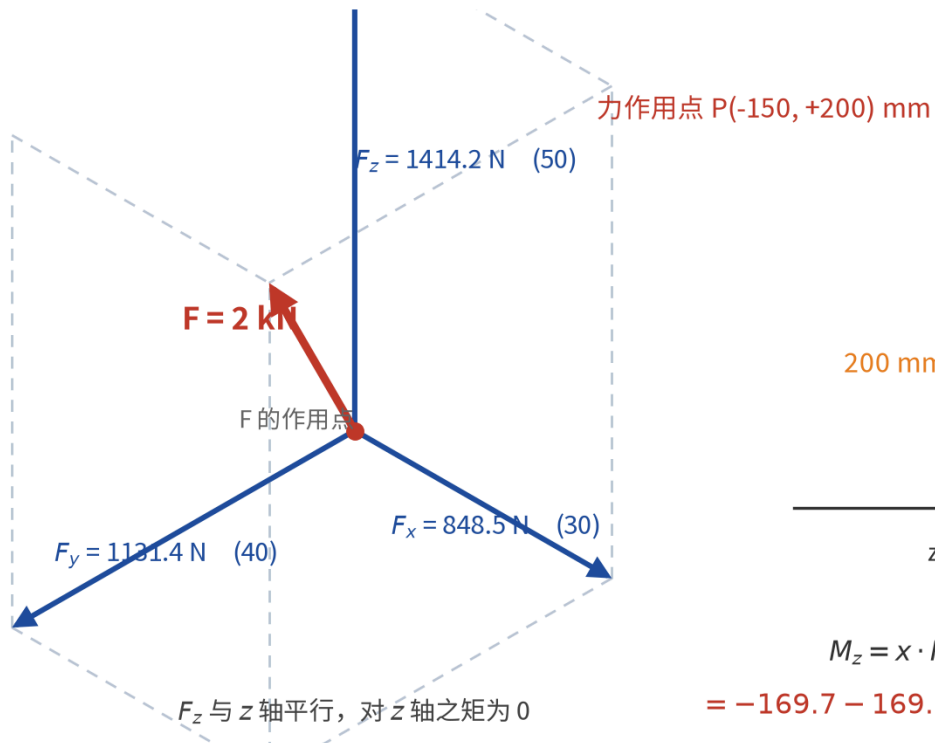
$$\Sigma F_x = 1229.0\text{ N}, \Sigma F_y = -119.0\text{ N}$$
$$R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2} = 1235\text{ N} \rightarrow 5.53^\circ \text{ (偏在轴线下方)}$$

(b) 力多边形：三力首尾相接，闭合矢量即 R

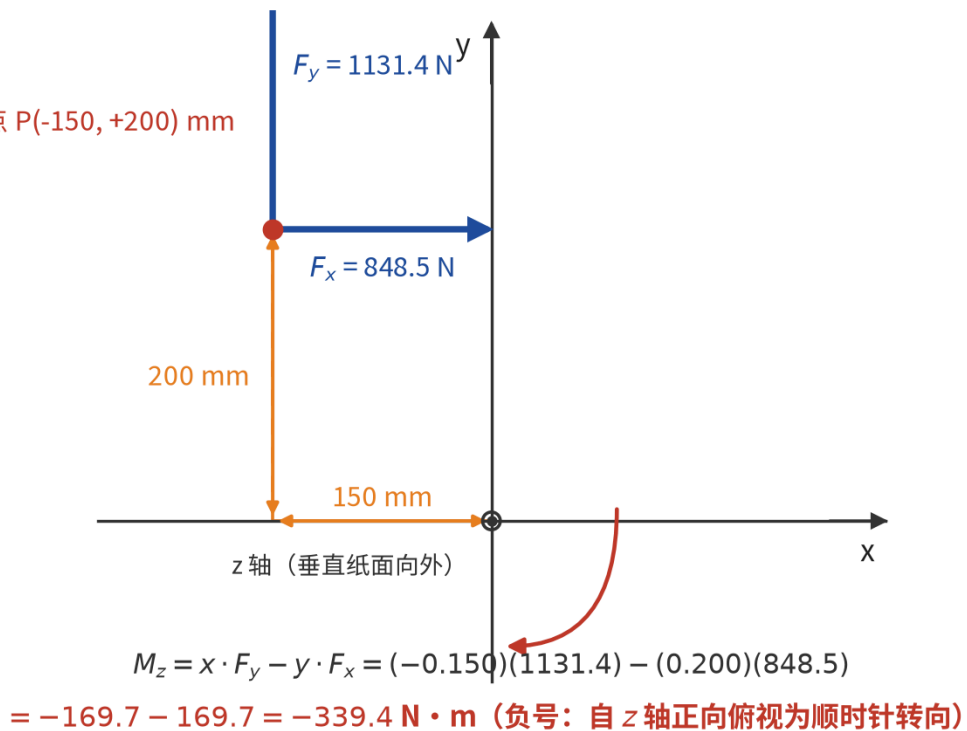


三力平移首尾相接后
起点 $\rightarrow$ 终点的闭合矢量就是合力 R

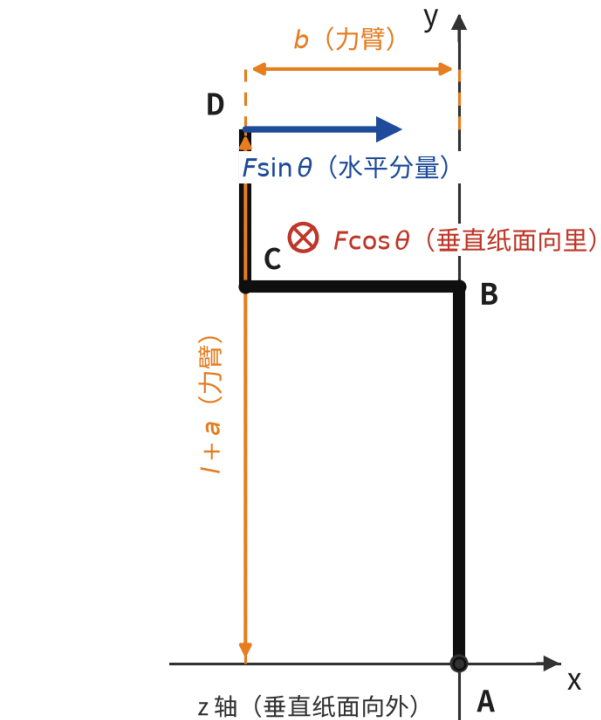
(a) F 沿 30:40:50 的长方体对角线分解



(b) 俯视图 (沿 z 轴向往下看): M_z 的两项



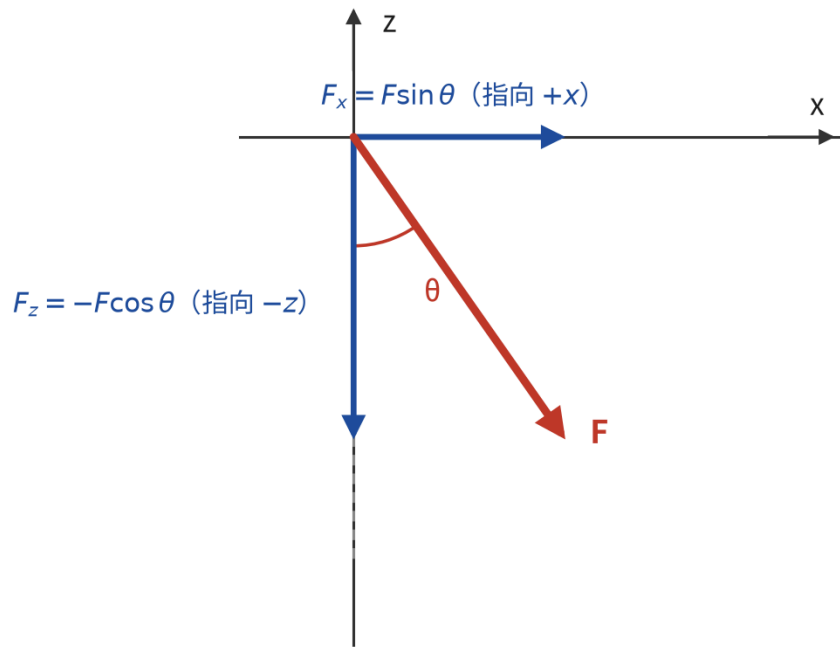
(a) 俯视图：直角曲柄 ABCD 与两个力臂



$$M_z = x \cdot F_y - y \cdot F_x = -(l+a) \cdot F\sin\theta$$

($F_y = 0$ ，故沿杆方向的力臂不起作用)

(b) xOz 平面：力 F 的分解



三个式子（力对轴之矩 = 坐标 $\times$ 力分量之差）：

$$M_x(F) = -F(l+a)\cos\theta$$

$$M_y(F) = -bF\cos\theta$$

$$M_z(F) = -F(l+a)\sin\theta$$